9.5 Making Inferences from Sample Surveys

Name _____ Period ____ Date _____

Algebra II 9.5 · Student sheet

I can…

- estimate a population mean or proportion from a sample.
- find and interpret the margin of error.
- use the margin of error to judge a claim.

Vocabulary · fill in the missing word

Descriptive statistics — numbers that describe a _____ set

Inferential statistics — use a sample to make _____ about a population

Margin of error — about $\pm\frac{1}{\sqrt{n}}$ for a _____ sample of size n

Interval — sample percent ± margin of error: likely to contain the _____ percent

Part 1 · Sample the bag

Draw 25 slips one at a time, putting each one back. Count the red slips.

Red slips	Sample size	Sample percent red	Margin of error ±1/√n	Interval
	25			

Class sample percents: _____

The bag is really 40% red. How many class samples had an interval that contained 40%? _____

Part 2 · Notes and examples (fill in each note, then work the example)

Example 1

A random sample of 8 students' study times (minutes): 30, 45, 20, 60, 35, 40, 25, 45. Estimate the mean study time of all the students.

Example 2

In a random sample of 250 voters, 140 support a measure. Estimate the percent of all voters who support it.

Margin of error: Margin of error ≈ ± _____

The interval: Interval = sample percent _____ margin of error

Example 3

In a random sample of 400 voters, 56% support a measure. Find the margin of error and the interval likely to contain the percent of all voters who support it.

Sample size: A larger sample makes the margin of error _____.

Example 4

A poll of 900 people finds that 47% support candidate A and 53% support candidate B. Can candidate B be confident of winning?

Part 3 · Partner practice

Partner 1

Find the margin of error for a sample of 1,600 people.

Partner 2

In a random sample of 100 students, 62% want a longer lunch. Find the interval likely to contain the percent of all students who want a longer lunch.